

KUMARA GURU.R

Process Engineer /Data Scientist

SUMMARY:

Process Engineer with a passion and experience in sustainability and Data Science.

A highly self-motivated Professional with strong interpersonal skills and a global mindset with passion for knowledge.

Experienced in delivering Projects on time with interdisciplinary department coordination.

Expertise in process optimization and sustainable innovation.

SOFTWARE SKILLS

- Aspen HYSYS
- Aspen Plus
- Aspen Edr
- HTRI
- AutoCAD
- Jupyter Notebook
- Pycharm
- Ms Office
- Github

TECHNICAL SKILLS

- Sustainability Analysis
- CO2 Emission
- Eco-Friendly Process
- Modelling and Optimization
- Chemical Process Calculation

PYHTON SKILLS

- Data Science /ML Modeling
- Multiple Chemical lib like
RDKit,ChemBL,
Pubchem,Chemdes,
Padelpy , Thermo ,
Chemicals, etc.
- Pandas
- Matplotlib
- Plotly
- Scikit-learn
- Tensorflow
- Keras
- PyQt5

+91 9790871497

r.kumaraguru44@gmail.com

Ambattur, Chennai ,Tamil Nadu - 600058.

<https://www.linkedin.com/in/kumaraguru-rajendran-95a456108>



EXPERIENCE

Simreka Softwares Pvt.ltd
Bengaluru

2022-Present

As Process Engineer / Data Scientist

- Backend Calculations script using Python.
- Process modelling using Simulation Software.
- Impact Calculation and Process Optimization to reduce the environmental impact of chemical and process plant.
- ML modelling for R&D development in chemical and Pharma plants.
- ML and DL models for Chemical Property prediction.
- Creating Thermodynamic packages using Physical and ML models for process plants.
- Creating Database of chemicals with their sustainability impacts.

V R Process Engineering
Consultants Pvt. Ltd
Chennai

2020 –2022

As Process engineer

- Study the complete process and perform laboratory testing to design the most suitable method for the better performance of the plant under the standard.
- Simualtion , Design , Process modeling and P&ID for chemical and Pharma plants.
- Estimating material, energy, water balance, and device an efficient way to reduce loss.
- Design Reactor, Heat Exchangers, and other equipment for Process plants.
- Optimization for Productivity enhancement under targeted Budget.

IDEAL Fasteners India Pvt. Ltd
Chennai

2019-2020

As a Quality Inspector and chemical analyst

- Chemical Testing for Hazardous Metal Contents.
- Developing new electroplating products.
- R&D in metal alloys used in wearables.

CERTIFICATIONS

- Aspen HYSYS
- Machine Learning
- Tensor Flow
- Data Science with Python
- Deep Learning

Academics

B.TECH

Anna university – BIT campus

2018

First class

Petrochemical Technology

Project

- Handled multiple projects for CO₂ Emission and Optimizing for Sustainability improvements.
- Experience in Designing ML and NN models for R&D purposes in Process Companies.
- Worked on Developing Back end calculation scripts in Python for Simreka Software.
- Developing sustainable alternatives for polymer and leather manufacturing companies.
- ML models for Synthesis and R&D purposes for companies.
- Process Modelling, Design and optimization for multiple Process plants.
- Developing Thermodynamic library incorporating both Physical and ML models.
- P&ID for process plants
- Handling Complete projects in Developing Eco-Friendly Processes such as Bio-Diesel Plants and Meat Waste Processing Plants.
- Experience in Solar power production .
- R&D experience in Alloy production and Electro Plating Materials.
- Experiences in Designing Biogas synthesis plant .
- Erection and commissioning of multiple Chemical Process plants.
- 20TPD Bio-diesel plant in Kerela.
- 35TPD Oleoresin Extraction Plant in Rajasthan .
- Bio-gas plant design for NTPC Jharkhand.
- Multiple Equipment(Heat Exchanges, Distillation Column etc.) designed for Various Process Plants.
- Chemical testing for harmful metal contents in wearable metal products

Personal Projects

- Build Personal Software for LCA Calculation using Pyqt5 for FrontEnd, Pandas, numpy and other Python libraries for calculation,matplotlib for graphs and SQLite for Database.
- Optimization, Design and operation of paraffinic crude sweetening plant through Aspen Hysys and Aspen EDR” -2nd price in National Conference (**NETCPT’17 -2017**).
- Simulation, design and optimization of steam reforming plant using Aspen Software(HYSYS, EDR & PLUS).

Reference

Dr.M.Rengasami 9443533238
Assistant professor
Anna university –Trichy

Mr.Ravindran 8608987912
Engineer
V R Process engineering
Consultants Pvt. Ltd

Mr.Prem Kumar .PhD 7418514424
Data Scientist
Simreka Softwares Pvt. Ltd

HOBBIES AND INTERESTS

- Chalk Sculpting
- Practice martial arts
- Drawing
- Watching movies



LANGUAGES

- Tamil
- English
- Can handle other language with Google translator .

Declaration:

I hereby declare that all the statements above are true, complete and correct to the best of my knowledge.

(Kumara Guru.R)